

Things I Don't Understand About Quantum Error Correction

A naive audit of the constructs — the confusions of a self-taught researcher, treated as data

CWI Summer School on Quantum Algorithms & Quantum Error Correction · 24–28 August 2026 ·
Poster Session: Wed 26 Aug, 16:30–18:00 · Rowan Quni-Gudzinis · Independent Researcher

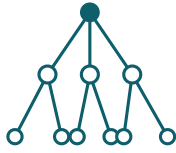
Panel 1 — Why this poster exists

I am an independent researcher, not a member of any quantum-error-correction group. I came to QEC from the outside — from ultrametric geometry and the energy cost of computation. On the way in, I collected **eight questions I cannot answer myself**.

This poster is my notebook made public. The questions are stated as questions — and my own first falsifiable prediction, log-periodic oscillations in the Planck 2018 temperature spectrum, came back **null** when I tested it (10.5281/zenodo.21902891). I hold my own ideas to the same standard I am about to ask you to hold them to.

The invitation: answer any of the eight questions and you will have taught me something real. Correct me, and the poster has done its job.

Panel 2 — The Method: feigned naivete as probe



I drew the structure I kept reading about — the Bruhat-Tits tree T_2 ($p=2$, 3-regular branching; the full construction truncates at depth 3, giving 22 vertices and 21 edges). Drawing a structure is when its assumptions become visible.

I ask the questions I'm not supposed to need to ask. The naive question is the **external structural probe**: it finds the locales and the seams. Where imports contradict, understanding breaks — **that break is data, not deficit**.

Method lineage: IAPS §6.2 audit-the-map; the Universal Ignorance Audit (15 questions, 10.5281/zenodo.21901984); "Knowing What We Do Not Know" (10.5281/zenodo.21901983).

Panel 3 — Things I don't understand about QEC

1 · Why the discreteness?

QEC protects a discrete abstraction (gates, qubits, circuits). Is the discreteness physical — or imported from the circuit model? If the abstraction is a choice, fault tolerance is the tax on the choice.

2 · What is a logical qubit, physically?

Hardware is bosonic oscillators with leakage levels; codes assume qubits. The map is 2D; the machine is continuous. Where does the map end?

3 · What does a threshold actually guarantee?

The proofs assume stochastic local noise. Coherent and correlated errors are the seams nobody prices. What is the guarantee outside the model?

4 · Where is the quantum in the loop?

Syndrome → classical decoder → feedback. QEC is quantum memory + classical control. Why is it called quantum error correction?

5 · Why isn't the decoder part of the code?

A classical algorithm decides a quantum code's performance. Coding theory and classical computation — two silos, never audited for compatibility.

Panel 3 (cont.) — The questions (2 of 2)

6 · Why 2D geometry?

Why a surface code and not a tree? Noise geometry is assumed continuous and Euclidean. What if error propagation is hierarchical — ultrametric (Bruhat-Tits)? The staircase nobody has looked for.

7 · Why count T gates, not joules?

The field prices the algebraic resource (T-count; magic states). Nobody prices the ancilla factory, verification retries, the decoder, the feedback. What does a correct answer cost in energy?

8 · Why is the cat the unit of protection?

Shor states are cat states — exponentially fragile to dephasing; verified ancillas go stale between verification and use. Why build protection from something that fragile?

Each question is a probe: the answer either closes the seam (the imports were compatible all along) or reveals it (the construct carried more than its proof).

Panel 4 — The seams: where the map ends

Seam	The contradiction
Discrete abstraction	Continuous physics → discrete gates → QEC defends the discreteness. The defense is priced; the choice is not.
Qubit / boson	The model is finite-dimensional; the hardware leaks. Every code assumes what the hardware denies.
Noise model	Stochastic-local assumptions meet coherent, correlated, adversarial reality. Thresholds live inside the model.
Classical control	The decoder is classical. The "quantum" loop runs on a classical brain — the field knows, and doesn't price it.
Cost currency	T-count is countable, so it is counted. Joules are measurable, and not measured. The algebraic unit stands in for the thermodynamic one.
Geometry	Euclidean 2D assumed; ultrametric tree unexamined. The staircase signature has never been searched for.

What I do understand: the maps are excellent at what they map. The seam is not the diagram — it is the **silence about where the diagram ends**.

Panel 5 — The statement, the caption, and the questions for the room

The one-line statement: "I cannot grasp quantum computing through the practitioners' constructs because the constructs are maps drawn by different cartographers — internally consistent, mutually contradictory. **My confusions are the fault lines.**"

The caption (feigned naivete is a method, not a pose): the naive question is the probe that finds the seam. I am not claiming the field is wrong — I am claiming the imports were never checked against each other. *If the constructs were compatible, the naive questions would answer themselves.*

Questions I will ask this week:

- "Is the discreteness of the circuit model physical or imported?"
- "Has anyone priced the energy of the decoder — per correct solution?"
- "Why a 2D cluster and not a tree?" (Bruhat-Tits resource states)

References (verified 2026-08-18; updated 2026-08-21)

Planck 2018 log-periodic null — 10.5281/zenodo.21902891

Universal Ignorance Audit — 10.5281/zenodo.21901984 · Knowing What We Do Not Know — 10.5281/zenodo.21901983

Ultrametric Code Spaces — 10.5281/zenodo.21824194 · JPCUB (joules-per-solution) — 10.5281/zenodo.21945415

Tree-code testing roadmap — QNFO/cwi-qec-poster-2026 · poster-v5-treecode

Rowan Quni-Gudzinis
Independent Researcher · Adelic Physics Programme
papers.qnfo.org · zenodo.org/communities/qnfo

Acknowledgment: CWI Summer School —
Research Semester Programme
"Quantum Algorithms & QEC"
(no external funding) · v4-noZX — naive audit, ZX anchor
removed, 2026-08-21

